

Networks are used to establish connections and route packets in and out of your local network. They pose the greatest danger, especially for the transferability of malware. Securing the networks by monitoring telemetry is vital to the success of our project.

Applications are what users mainly interact with; these applications do a vast number of functionalities to ensure proper working of their software. As well as if the user wants to host a web server and database. It is sometimes the first source of entry, especially if the application itself is malicious or used to be the medium of malice. Application telemetry is important as it may serve as the first line of defense.

Network Telemetry:

Event	Where it comes from	Why it matters	What looks suspicious
New port opening	/proc/net/tcp, ss -tulpn	What services are waiting to accept external connections	Random unknown port number, high port numbers, unexpected port numbers
Outbound connections in use	/proc/net/tcp, ss -ntup	Where this machine is calling out to	Unknown IPs addresses, known malicious IPs
Promiscuous Mode Enabled	/var/log/syslog	Indicates all traffic interception	"[nic] entered promiscuous mode" w/o actively using sniffing tools
Change in Firewall Rules	/var/log/ufw.log	Control over firewall persists	A suddenly disabled firewall rule or addition
DNS requests	Systemd-resolved logs, /var/log/syslog	Shows which domain names system is trying to resolve	Bad domains that are known to carry malware, or unknown unreasonably large domain names
ARP table changes	/proc/net/arp	ARP spoofing	Duplicate MACs, IPs, and unknown MACs
SSH tunneling and port forwarding	/proc/net/tcp, lsof, /var/log/auth.log	Used to pivot via SSH	Local/remote forwards, unexpected sshd children

Large Data Transfer	/proc/net/dev, iftop	Data exfiltration	Sudden spike to unknown IPs (external)
SMB/NFS mounts	/proc/mounts, proc/net/dev	Unauthorized sharing of networks	New remote mount not included in baseline
Raw Socket Use	/proc/net/raw	Used by attackers to create raw packet crafting	Unknown or unusual processes using raw sockets
VPN tunnel creation	proc/net/dev	Use of unauthorized tunnels can be used to exfiltrate data	Creation of tunnel interface, unusual GRE/IPsec traffic

Application Telemetry:

Events	Where it comes from	Why it matters	What looks suspicious
SSH key Modification	~/.ssh/authorized_keys	Modifying SSH keys is a step used by attackers to establish persistence	New key, deleted key
Abnormal Server Requests	/var/log/nginx/access.log	If deploying a web server, ensuring traffic requests are normal is vital to application security	Many 404 errors,
Configuration changes	/etc/	Changing configurations can create backdoors and/or weaken security	
Shell History	~/.bash_history	Shell history can be used by attackers to	Wiping shell history, symlinking to

		limit the amount of information can be gathered after an attack	>dev/null
Database Errors	/var/log/mysql/error.log	If hosting a database, attackers may target critical data	Repeated local authentication attempts to database
Malicious Macros	/proc/<pid>/stat	Documents should never execute commands (system)	LibraOffice spawning bash, wget, curl
Browser Credentials	~/.mozilla/ or ~/.config/google-chrome/ or ~/.config/microsoft-edge/	Attackers often target credentials that are saved in the browser	Script copying the data file containing credentials
Autostart Persistence (User)	~/.bashrc or ~/.config/autostart	Use things to automatically execute certain malware every time a user logs on.	New file inserted into autostart folder (especially .desktop), code being appended to .bashrc
Sus Downloads	~/Downloads	Drive-by downloads or phishing material land here first	Siles ending in .sh or .elf and is franted execute permissions instantly
Microphone/Camera activation	/var/log/syslog	User want to know if their microphones or cameras were used without permission	A background process with no GUI requesting access to /dev/video0 or /dev/snd (captured audio)
Cron Job Changes	/etc/cron*, /var/spool/cron/crontabs	Used to establish persistence	Unusual cronjobs with unkown commands or very unusual intervals
Browser Extension Change	~/.config/google-chrome/Default/Extensions, ~/.config/microsoft-edge/Default/Extensions/ ~/.mozilla/firefox/[ProfileID].default-release/extensions/	Can be used to install malicious extensions	New extension not installed by user

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